

The Respiratory System

Moving air is mechanics. Moving gas is diffusion. They are two different problems.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- Explain how a pressure difference moves air in and out
- Say what surfactant does, and why a premature baby struggles without it
- Explain gas exchange using pressure differences
- Describe how oxygen and carbon dioxide each travel in blood
- Read the oxygen curve and predict what shifts it

1 Two zones, two jobs

Most of the airway does no gas exchange at all.

The **conducting zone** is the nose, windpipe, and branching tubes. It warms, moistens, and filters air, then delivers it. No exchange happens here, which is why it is called dead space.

The **respiratory zone** is the air sacs. Every bit of gas exchange happens there.

That surface is huge and the barrier is thin: one air sac cell and one capillary cell. Anything that thickens the barrier, like fluid, infection, or scarring, slows diffusion and shows up as shortness of breath.

2 Breathing is a pressure problem

- 1 **The diaphragm contracts and flattens**, and the ribs lift. The chest gets bigger.
- 2 **Bigger space means lower pressure** inside the lungs, now below the pressure outside.

- 3 **Air flows in**, because gas moves from higher pressure to lower. That is all breathing in is.
- 4 **The diaphragm relaxes**, the chest springs back, space shrinks, and pressure rises.
- 5 **Air flows out**. Quiet breathing out is passive, so it costs nothing.

The lungs do not pull air in. They are inflated by a pressure difference the chest wall creates around them, and they stay stuck to that chest wall because of the sealed space between them.

WHY THE NURSE CARES

Break that seal with a chest wound and air gets into the space. The lung is no longer held open against the chest wall, so it collapses. Nothing is wrong with the lung itself. The thing holding it open is gone.

Surfactant is why small air sacs stay open. A wet surface pulls itself closed, and the smaller the bubble, the harder it pulls. Surfactant reduces that pull, so air sacs do not collapse at the end of each breath and do not have to be reopened from scratch. Premature babies may not make enough yet, which is exactly why their breathing is so hard.

3 Gas exchange costs no energy at all

Every gas moves from where there is more of it to where there is less. Nothing pumps it.

WHERE	OXYGEN	CARBON DIOXIDE	NET MOVEMENT
Air sac vs arriving blood	High in the air sac	High in the blood	Oxygen into blood, carbon dioxide into the air sac
Blood vs working tissue	High in the blood	High in the tissue	Oxygen into tissue, carbon dioxide into blood

It is the same rule in both places. The direction flips only because the difference flips. Learn the rule and you never have to memorize the direction.

4 How each gas actually travels

- **Oxygen.** About 98% is carried on hemoglobin, and the rest is dissolved. The dissolved part is small, but it is what the pressure reading measures, and it is what drives loading and unloading.
- **Carbon dioxide.** Most of it is turned into bicarbonate inside the red cell. Some rides on hemoglobin, and a little is dissolved.

That bicarbonate route links breathing straight to acid balance. Carbon dioxide plus water makes an acid, which splits into bicarbonate and a hydrogen ion.

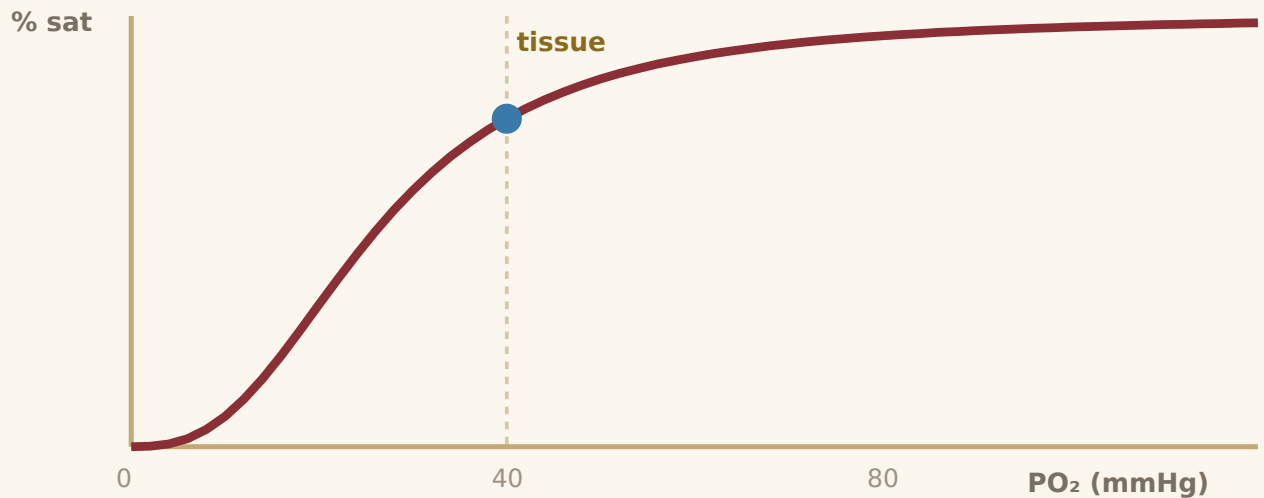
So holding on to carbon dioxide builds up hydrogen ions and makes the blood acidic. Blowing carbon dioxide off does the opposite.

That is why a patient who breathes too little becomes acidic, why one who over-breathes becomes alkaline, and why the lungs can correct a body-wide acid problem in minutes just by changing how hard the patient breathes.

5 The curve, and why it bends

Move the tissue conditions and watch how much oxygen gets released.

OXYGEN-HAEMOGLOBIN DISSOCIATION



Resting conditions. At a tissue PO₂ of 40 mmHg haemoglobin is about 76% saturated, so roughly a quarter of the oxygen carried is handed over on each pass — leaving a large reserve for when demand rises.

The curve is S-shaped for a reason.

The **flat top** means that in the lungs, oxygen loading stays high even if the oxygen in the air drops somewhat. That is a built-in safety margin.

The **steep bottom** means that in the tissues, a small drop in oxygen releases a lot of it.

The curve also **shifts**. Working tissue is hot, acidic, and full of carbon dioxide, and every one of those makes hemoglobin let go of oxygen more easily.

So the tissue that needs oxygen most changes its own local conditions in the way that delivers it. There is no signal and no control center. The chemistry does it.

THINK OF IT THIS WAY

Hemoglobin holds oxygen loosely in a hot, acidic crowd and tightly in a cool, calm one. Exercising muscle makes itself exactly the kind of place where hemoglobin lets go.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. What directly causes air to flow into the lungs?

- a) The lungs expand and pull air in
- b) The chest gets bigger, so pressure inside the lungs drops below the pressure outside
- c) The diaphragm pushes air downward
- d) The air sacs squeeze to draw air in

2. A chest wound lets air into the space around the lung, and the lung collapses. Why?

- a) The lung tissue was destroyed
- b) The pressure difference holding the lung open is gone
- c) The airway is blocked
- d) Surfactant washed away

3. Why does a premature baby without enough surfactant struggle to breathe?

- a) The airways are too narrow
- b) Air sacs collapse between breaths, so each breath must reopen them
- c) The diaphragm is not connected yet
- d) There is no hemoglobin to carry oxygen

4. A patient is breathing too slowly and shallowly. What happens to blood pH?

- a) It rises, so the blood becomes alkaline
- b) It falls, because held-in carbon dioxide makes hydrogen ions
- c) It does not change, because only the kidneys set pH
- d) It rises, because oxygen levels fall

5. Working muscle is hot, acidic, and full of carbon dioxide. What does that do to oxygen delivery?

- a)** Hemoglobin holds oxygen tighter, protecting the supply
- b)** Hemoglobin lets go of oxygen more easily, so more is delivered
- c)** Nothing changes
- d)** Oxygen switches to travelling in plasma

Answer key

1. **The chest gets bigger, so pressure inside the lungs drops below the pressure outside**

Correct. Bigger space means lower pressure, and gas flows toward lower pressure.

2. **The pressure difference holding the lung open is gone**

Correct. The seal is what kept the lung open. Break it and the lung springs back.

3. **Air sacs collapse between breaths, so each breath must reopen them**

Correct. Reopening collapsed air sacs on every breath takes enormous effort.

4. **It falls, because held-in carbon dioxide makes hydrogen ions**

Correct. Carbon dioxide and water make an acid, which releases hydrogen ions.

5. **Hemoglobin lets go of oxygen more easily, so more is delivered**

Correct. The tissue that needs oxygen most creates the conditions that release it.

BIO 262 • Human Physiology

Joyce University of Nursing and Health Sciences

Course resources by **Dr. Vidya Garg**